

Section A

Instructions:

1. Answer ALL questions in this section.
2. Where necessary use the following information:
 ASCII value for uppercase letter A is 065
 ASCII value for Lowercase letter a is 097

Q1. (a) Briefly explain what is meant by expression as used in C++ programming language and give two examples of expressions. **1 mark**

(b) Distinguish between an integral and float expressions giving two examples on each. **1 mark**

Q2. (a) What is the use of an assignment operator as in C++ programming language? **1 mark**

(b) Carefully examine the variable declarations and assignments in the given code block and determine the output.

```
{
    int x, c;          float y;          char ch;
    x = 10 + 3.142857;  y = 55.0 / 7;
    c = 'D';           ch = 65;

    cout << " The output is: " << endl;
    cout << " The value of x is " << x << endl;
    cout << " The value of y is " << y << endl;
    cout << " The value of c is " << c << endl;
    cout << " The value of ch is " << ch << endl;
}
```

1 mark

Q3. Using the standard library functions **pow(x, b)**, and/or **sqrt(x)** write a C++ expression corresponding to the following mathematical expressions.

(a) $y = \frac{3x^4 + 5x^3}{3x^2 + 6}$ **1 mark**

(b) $r = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$ **1 mark**

- Q4. (a) Mention four attributes that must be indicated in a function declaration? **1 mark**
(b) Give any four examples of a function declaration. **1 mark**

Q5 Determine the output that will be generated by the C++ codes in (a) and (b) below.

```
(a) {  
    int count, num;  
    num = 0;  
    cout << "The output is:" << endl;  
    for (count = num; count <= 30; count++)  
        if (count % 4 == 0)  
            cout << count << "  ";  
}
```

1 mark

```
(b) {  
    int angle, count;  
    angle = 0;  
    count = 0;  
    cout << "The output is: " << endl;  
    while(angle <= 40)  
    {  
        cout << angle << endl;  
        angle += 8;  
        count++;  
    }  
    cout << count << " angles generated.";  
}
```

1 mark

SECTION B

Answer ONE (1) question from this section

Q6. The moon gravity is about 17% of the Earth's gravity. Write a C++ program that allows you to enter your weight and computes your effective weight on the moon. Use a function other than **main ()** function dedicated for the computation. **10 marks**

Q7. Write a C++ program that compute the volume of a cylinder using two functions **main ()** and **CylinderVol ()**. Let the **main ()** program prompt the user to enter dimension values and pass these values to **CylinderVol ()** which perform the computation and return the result to **main ()** for the result display.

(Hint: Use the mathematical model $V = \pi r^2 h$).

10 marks

MBEYA UNIVERSITY OF SCIENCE AND TECHNOLOGY



ICT DEPARTMENT

END OF COURSE EXAMINATION, JUNE 2015/2016

COB 2202, GSB 2205 COMPUTER PROGRAMMING II

Date: 12TH July, 2016

Time: 7:45 - 10:45 (3 Hours)

Venue: SPORTS HALL

Instructions:

1. This paper consists of 4 printed pages with two sections A and B.
2. Section A carries 24 marks; while section B carries 36 marks.
3. Answer **ALL** questions in section A and any **THREE** questions from section B.
4. Do not write on the question paper. All rough work must be done at the end of the answer booklet and crossed through.

SECTION A (24 marks)

Attempt ALL questions in this section.

Question 1

- ✓(a) Distinguish between assignment and equality operators as used in C++ programming language? **2 marks**

- (b) Examine carefully the following variable declarations and assignments:

```
int x,;    x = 10 + 3.142857;
```

```
float y, z; y = 22.0 / 7;
```

```
char ch;   ch = 65;
```

```
z = y + 5;
```

Determine the values assigned to the variables x, y, ch and z.

2 marks

- (c) With an example distinguish between integral and float expression

2 marks

Question 2

- (a) Explain what is meant by a **control statement** and give two examples.

2 marks

- (b) Determine the output that will be generated by the C++ codes in (i) and (ii) below.

```
(i)  {
      int count, num;
      num = 0;
      cout << "The output is:" << endl;
      for (count = num; count <= 30; count++)
          if (count % 4 == 0)
              cout << count << " ";
  }
```

2 marks

```
(ii) {
      int angle, count;
      mangle = 0;
      count = 0;
      cout << "The output is: " << endl;
      while(angle <= 40)
      {
          cout << angle << endl;
          angle += 8;
          count++;
      }
      cout << count << " angles generated.";
  }
```

2 marks

Question 3

Using the standard library functions **sqrt()** and **pow()**, write a C++ expression corresponding to the following mathematical expressions.

Hint: **sqrt(x)** returns a non-negative square root of x, and **pow(x, y)** returns x^y .

a) $5x^3 + 10x^2 + 6x = 0$

2 marks

b) $z = \frac{3x^4 + 5x^2}{3x + y}$

2 marks

c) $\sqrt{\frac{2x^2 + 3y}{3x^3 - 6}}$

2 marks

Question 4

(a) Distinguish between **if-else** and **switch-case** statements. (2 marks)

(b) Convert the following **if-else** statements into **switch-case** equivalence.

```
if (ch == 'A')
{
    cout << a + b;
    cout << "Addition done!";
}
else if (ch == 'S')
{
    cout << , a - b;
    cout << "Subtraction done!";
}
else if (ch == 'M')
{
    cout << , a * b;
    cout << "Multiplication done!";
}
else if (ch == 'D' && b != 0)
{
    cout << a / b;
    cout << "Division done!";
}
else
    cout << "Wrong Choice!";
```

4 marks

SECTION B (36 marks)

Attempt any **THREE** questions in this section.

Question 5

A C++ program contains the following declarations and initial assignments as follows:

```
int j = 0, n = -1, p = 2;
unsigned int m = 1;
float x = 0.005, y = -0.01;
```

Determine the value of each of the following:

- a) $x += (y \geq n)$
- b) $x += y -= p$
- c) $m /= x * n + y$
- d) $(j \parallel m) + (x \parallel ++n)$
- e) $j \geq 10 \ \&\& \ x \geq 1 \ \&\& \ m$
- f) $++j == m != y * 2$

12 marks

Question 6

Write a C++ program that uses two functions other than **main()** to compute the area and volume of a cylinder. Name the functions as **cylinder_area()** and **cylinder_vol()**. Let the **main()** program prompt receive the radius and height of a cylinder, pass these values to the corresponding functions which perform computations and return the results to main for display.

(Hint: Use the mathematical models $A = 2\pi r^2 + 2\pi rh$, $V = \pi r^2 h$) 12 marks

Question 7

Using **if** or **if ... else** conditional statement, write a C++ program to convert temperature from degrees Fahrenheit to degrees Centigrade or vice versa. Provide the menu from which a user can make a choice of what a conversion to perform and let your program print the message "Wrong Choice!", when the user enters a wrong choice. (Hint: Use the mathematical model $C = (5/9)(F - 32)$).

12 marks

Question 8

Using **switch-case** statements write a C++ program that compute the area of a circle, the area of a rectangle and the area of a triangle. Let the program prompt the user to enter dimensions, perform calculations and print the result in each case.

(12 marks)

MBEYA UNIVERSITY OF SCIENCE AND TECHNOLOGY

COLLEGE OF ENGINEERING AND TECHNOLOGY



**DEPARTMENT OF INFORMATION AND COMMUNICATION
TECHNOLOGY**

SEMESTER II EXAMINATIONS 2016/2017

**GSB2205 - COMPUTER PROGRAMMING II
COB2202 - OBJECT ORIENTED PROGRAMMING**

UQF 7A

TIME : 11:00 AM – 2:00 PM (3:00 Hours)

DATE: 04/07/2017

INSTRUCTIONS:

1. This paper consists of 4 printed pages with two sections A and B.
2. Section A carries 24 marks; while section B carries 36 marks.
3. Answer ALL questions in section A and any THREE questions from section B.
4. Do not write on the question paper. All rough work must be done at the end of the answer booklet and crossed through.

SECTION A (24 marks)

Attempt **ALL** questions in this section.

✓Q1. (a) Define the following terms and give an example on each.

- (i) **Preprocessor directive** as used in C++ Programming language? **2 mark**
- (ii) **Input streams** and **output streams** in C++ programming language? **2 marks**

(b) What do you understand by the term **control statement** as used in high level programming language and give two examples? **2 marks**

✓Q.2 (a) With an example distinguish between **Built-in** and **Derived** data types giving two examples on each **2 marks**

(b) Briefly explain the meaning of the following giving an example on each.

- (i) Built-in functions **2 marks**
- (ii) User defined functions **2 marks**

Q.3 Using the standard library functions **pow(x, b)**, and/or **sqrt(x)** write a C++ expression corresponding to the following mathematical expressions.

(a)
$$y = \frac{3x^4 + 5x^3}{3x^2 + 6x}$$
 2 marks

(b)
$$r = \sqrt{\frac{2x^2 + 3y}{3x^3 - 5y}}$$
 2 marks

(c)
$$z = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$
 2 marks

Q4. (a) What do you understand by the term control statement? 2 marks

(b) Determine the output that will be generated by the following C codes.

```
(i) cout << "The output is:" << endl;
    for (num = 1; num <= 20; num++)
        if (num % 3 == 0)
            cout << num + 1 << "\t";
```

1 mark

```
(ii) int digit = 1;
    Cout << "The digits are: << endl");
    do
    {
        Cout << "\n" << digit;
        digit += 4;
    } while(digit <= 21);
```

1 mark

(c) Write a C++ program using a **for loop**, that prints the numbers between 30 and 100 and are evenly divided by 3. 2 marks

SECTION B (36 marks)

Answer any **THREE** questions from this section.

✓ Q5. Given the following declarations

```
int j = 0, m = 1, n = -1;
float x = 2.5, y = 0.0;
```

Evaluate the following expressions.

- i. $-x + j == y > n >= m$
- ii. $x += (y >= n)$
- iii. $++j == m != y * 2$
- iv. $!x || !n || m + n$
- v. $(x > y) || !j || n++$
- vi. $(j || m) + (x || ++n)$

12 marks

Q6. Using `if ... else` conditional statement, write a C++ program that convert temperature from degrees Centigrade to degrees Fahrenheit and vice versa. Provide the menu from which a user can make a choice.

✧ (Hint: Use the mathematical model $C = (5/9)(F - 32)$). 12 marks

Q7. Using `switch...case` statements write a C++ program to calculate the areas of a **circle**, a **triangle**, and a **rectangle**. Let the program provide the menu from which a user can make a choice. Let your program prompt the user to enter dimensions in each case.

✧ 12 marks

Q8. Write a C++ program that compute the volume of a cylinder using two functions `main()` and `CylinderVol()`. Let the `main()` program prompt the user to enter dimension values and pass these values to `CylinderVol()` which perform the computation and return the result to `main()` for the result display.

✧ (Hint: Use the mathematical model $V = \pi r^2 h$). 12 marks

MBEYA UNIVERSITY OF SCIENCE AND TECHNOLOGY

GSB 2205, COMPUTER PROGRAMMING II

TEST 2

TIME: 1:30 HRS

Date: 18th May, 2018

Instructions:

1. Answer question FOUR and ANY OTHER THREE questions.
2. Where necessary use the following information:
ASCII value for Uppercase letter is A is 065 and lower case a is 097

Q1. (a) Distinguish between Built-in and user-defined functions. 1 mark

(b) Describe any two types of user-defined functions and give an example of each. 4 marks

Q2. (a) Carefully examine the variable declarations and assignments in the given code block and write down the output that can be generated.

```
(  
    int x, y; float z; char ch;  
    x = 5; y = pow(x, 5); z = 10 + 3.142857; ch = 'p';  
    cout << "The output is: " << endl;  
    cout << "The value of y is " << y << endl;  
    cout << "The value of z is " << z << endl;  
    cout << "The value of z + (int)z is " << z + (int)z;  
    cout << "\nThe value of ch is " << ch << endl;  
    cout << "The value of y + ch is " << y + ch << endl;  
)
```

5 marks

Q3. (a) What are necessary preprocessor directives (header files), when the `exit()` and `pow()` library functions are to be used in a program: 2 marks

(b) Using the standard library functions `pow(x, b)`, and/or `sqrt(x)` write a C++ expression corresponding to the following mathematical expressions.

(i) $y = \frac{3x^4 + 5x^3}{3x^2 + 6}$ 1.5 marks

(ii) $r = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$ 1.5 marks

Q4. Write a C++ program that compute the volume of a cylinder using two functions `main()` and `CylinderVol()`. Let the `main()` program prompt the user to enter dimension values and pass these values to `CylinderVol()` which perform the computation and return the result to `main()` and display the result in the following format: **The volume of a cylinder is xx cubic units.**

(Note: Use the mathematical model $V = \pi r^2 h$

Qualify any constant used in your program.

xx stands for the calculated value of cylinder volume.)

5 marks

Q5. Using `if ... else ... if` ladder conditional statements write a C++ program to convert temperatures from degrees Fahrenheit to degrees Centigrade or vice versa. In your program provide the menu from what a user can make a choice of what conversion to perform. You may use the formula $C = (5/9)(F - 32)$. 5 marks

Section A

Instructions:

Answer ALL questions in this section.

Q1. (a) What do you understand by the term **expression** as used in high level programming language and give two examples? **2 marks**

(b) Identify **any two** types of expressions giving an example on each. **2 marks**

Q2. (a) What is the use of an assignment operator in C++ programming language? **1 mark**

(b) Examine carefully the following variable declarations and assignments:

```
int x;  
x = 10 + 3.142857;  
float y;  
y = 22.0 / 7;  
char ch;  
ch = 65;
```

Determine the values assigned to the variables x, y and ch. **3 marks**

Q3. (a) Distinguish between a Built-in function and User-defined function and give an example of each. **2 marks**

(b) Identify the attributes associated with a function prototype. **2 marks**

Q4. Using the standard library functions **sqrt(x)** and **pow(x, b)**, write a C++ expression corresponding to the following mathematical expressions.

(a)
$$z = \frac{3x^2 - yz + 4}{5x^3 - 6x + 1}$$
 2 marks

(b)
$$r = \frac{+b - \sqrt{b^2 - 4ac}}{2a}$$
 2 marks

Q5. Determine the output that will be generated by the C++ codes in (a) and (b) below.

(a) {

```
int count, num;
num = 49;
cout << "The output is:" << endl;
for (count = num; count >= 0; count--)
if (count % 7 == 0)
    cout << count << " ";
cout << "\nCont from " << num << " to " << "0";
```

2 marks

(b) {

```
int angle;
angle = 10;
cout << "The output is:" << endl;
do
{
    cout << angle << "\n";
    angle = angle + 30;
} while (angle <= 200);
```

2 marks

Section B:

Answer **ONLY ONE** (1) question from this section.

Q6. Write a C++ program that uses a function called **cylinder_vol()** to compute the volume of a cylinder. Let the **main()** program prompt the user to enter the radius and height of a cylinder, pass these values to **cylinder_vol()** which perform the computation and return the result to **main()** for display. (Hint: Use the mathematical model $v = \pi r^2 h$).

10 marks

Q7. Using **switch-case** statements write a program that compute the area of a **circle**, the area of a **rectangle** and the area of a **triangle**. Let the program prompt the user to enter dimensions, perform calculations and print the result in each case. For simplicity assume C++ is not case sensitive.

10 marks



GSB 2205, COB 2202 Object Oriented Programming

TEST 1

Date: April 21st, 2017

TIME 1:15 Hours

INSTRUCTIONS:

1. Answer ALL questions
2. Do not copy questions. Just write the answers/solutions.

Q1. Write **T** for a TRUE statement and **F** for a FALSE statement on each of the following arguments. @ 0.5 marks

- (i) C++ introduces object oriented concepts over C programming language. **T**
- (ii) Statements for creating data storage in a C++ function body are called local variables. **T**
- (iii) New features of C++ programming language include format free input and output expressions. **T**
- (iv) The extension .cpp on a C++ file name enables the compiler to include the needed standard library codes at the compilation time. **T**
- (v) The preprocessor directive #include<iostream.h> contains instructions and predefined constants that are used in the program. **T**
- (vi) The function declarator in a C++ program marks where the execution of the program begins. **T**
- (vii) The body of the function in a C++ program must be enclosed between "(" and ")" brackets. **F**
- (viii) The word main() in C++ programming language is a keyword word. **F**
- (ix) Linking the code in C++ programming language means combining all the required codes in the program and result start-up code. **F**
- (x) C++ streams do not require explicit data types specification in I/O statement but requires explicit address operator prior to the variable in the input statement. **F**

(5 marks)

Q2. (a) (i) Briefly explain what meant by linking the code in C++ programming language. **1 mark**

(ii) What are the uses of the input and output streams in C++ programming. **1 mark**

linking - process of combining all required object and start up code to perform
 input -> Allow read operations with input device eg disk, keyboard
 output -> Allow write operations on the output device.

- (b) (i) With an example distinguish between an **operator** and **operand** as used in C++ programming. **1 mark**

(ii) Given the declarations

`int j = 0, m = 1, n = -1; float 2.5, y = 0.0; x = 2.5`

Evaluate the following expressions. @ 1 mark

• `-x + j == y > n > m`

1 mark

• `x += (y >= n)`

1 mark

- Q3.** Assume you are running the given the c++ code block; clearly indicate the responses to the prompt and the output generated.

```
{
char FirstName[12], SecondName[12];
int age;
char *my_address = "P. O. Box 131, Mbeya";
long int PhoneNo;
float PhoneBill = 25000.50;
cout << "Enter your first name: ";
cin >> FirstName;
cout << "Enter your second name: ";
cin >> SecondName;
cout << "Enter your phone number: ";
cin >> PhoneNo;
cout << "\n\n";
cout << "The output is:" << endl << endl;
cout << "Name: " << FirstName << " " << SecondName;
cout << "\n" << "Address: " << my_address << endl;
cout << "Phone number: " << PhoneNo;
cout << " " << "Phone Bill: " << PhoneBill << endl;
cout << "City: ";
cout << " MBEYA";
}
```

5 marks

- Q4.** Write a C++ program that compute the area of a cylinder. Let your program prompt the user to enter the radius and height of a cylinder performs computation and output the result. Make use of constant qualifier for any standard constant used in your computation.

Hint: use the mathematical model $A = 2\pi r^2 + 2\pi rh$

5 marks

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